

Suiyang Guo

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Control Eng. | 2024.09—2027.06



MSc student in Control Engineering at Northwestern Polytechnical University (NPU), specializing in robotics and AI (graduating Jun 2027). Full-stack robotics engineer with end-to-end capability from mechanical/PCB design and STM32 firmware to ROS/ROS2, deep learning and host software. Currently a VLA (Vision-Language-Action) & World-Model algorithm intern at Visincept (incubated by IDEA Research / International Digital Economy Academy), training and deploying manipulation policies for a dual-arm robot. Participated in a national key research project; holds 6 patents (4 invention, 2 utility-model) and 2 software copyrights.

Education

Northwestern Polytechnical University (NPU)	2024.09—2027.06
MSc, Control Engineering Robotics & AI GPA: 89.56/100	
Zhengzhou University	2020.09—2024.06
BEng, Automation (Elite Class) Robotics & AI GPA: 3.54/4.0	

Technical Skills

- **AI / Embodied AI:** VLA models ($\pi 0.5$, X-VLA) training & real-robot deployment, flow-matching policy, action chunking, RTC real-time inference, latent-milestone world models; PyTorch (DDP) / JAX-Flax (FSDP) distributed training
- **Deep Learning:** end-to-end training in PyTorch / JAX, DINOv3 / SigLIP visual representations, CNNs, speech recognition; Triton kernels & CUDA Graphs for inference acceleration
- **Programming:** C/C++ (proficient), Python (proficient), MATLAB (fluent); Linux driver/application development
- **Robotics:** ROS/ROS2 (proficient); navigation, SLAM, motion planning, inverse kinematics, point-cloud processing
- **Embedded & Hardware:** STM32/51/Arduino (proficient; IMX6ULL/STM32F4), FreeRTOS; PCB design (Altium Designer, JLCPCB), 3D printing
- **Tools:** Vibecoding (AI-assisted / agentic coding with Claude Code etc.), Git, PyQt5 (host software), Autodesk Inventor (3D modeling)

Experience

Visincept (incubated by IDEA Research)	2026.04—Present
VLA & World-Model Algorithm Intern	
• Trained a dual-arm cloth-folding policy on the open-source $\pi 0.5$ (openpi) VLA; a curated-data + two-stage-init recipe cut action error (offline MAE@50) by about 77% vs. the prior best, and shipped to real-robot deployment • Led VLA inference real-time optimization (RTC real-time chunking + Triton kernel autotuning), cutting single-step inference latency from about 256 ms to 32 ms (roughly 8x) and end-effector jitter by 81%; drove the ROS1 to ROS2 migration for on-robot serving • Built an AWBC / RECAP-advantage offline policy-improvement pipeline; ran cross-embodiment co-training and weight-space model-merging experiments • Brought up Tsinghua's X-VLA architecture on our platform, using firmware-level IK to cut the control cycle from 1.4 s to 0.33 s; built an offline vision-ablation gate that root-caused and fixed a dataloader bug which had made the model vision-blind • Explored a self-supervised latent-milestone world model (LMVLA), injecting predicted sub-goals into the $\pi 0.5$ action expert; evaluated on LIBERO / RoboTwin	
Zhongqin Herui (Shaanxi) Technology Co., Ltd.	2025.11—2026.01
Navigation & Mapping Algorithm Engineer (Part-time)	
• Developed 3D mapping and navigation for an orchard autonomous inspection vehicle • Owned equipment/sensor selection, calibration, and sensor-data processing • Built 3D maps from raw LiDAR, then occupancy grids for vehicle navigation	
Huawei Digital Energy Bootcamp	2025.07
Trainee (Training)	

- Studied digital-energy cloud microservice (DDD) architecture and AI applications/engineering in energy services
- Practiced prompt engineering and DevOps deployment workflows

Dongfeng Motor Co., Ltd.
Study Visit

2023.08

- Studied gasoline/diesel engine structure and principles; toured the full vehicle manufacturing process

Projects

Series-Parallel Hybrid Wheeled Humanoid Robot
National Key Research Project | Key Provincial Lab —team member

2024.09—Present

- **Overview:** designed and built a series-parallel hybrid wheeled humanoid system —AGV base, parallel mechanism, two 6-DoF arms (in-house 6-axis / JAKA) and a 2-DoF head —for agricultural picking, teaching and industrial use
- **Stack:** Ubuntu 20.04, ROS Noetic, STM32F4, point-cloud processing, path planning, ShenTong DB
- **Work:** wrote the robot master-control program (3D mapping, path planning, object recognition, dynamic obstacle avoidance); point-cloud publishing, collision detection, IK; designed an in-house multi-DoF arm control scheme, doubling (100%) the 6-DoF arm control frequency; STM32F4 lower-level control; built a ShenTong database for high-concurrency CRUD

AI-LLM Voice-Interaction Robot
Outstanding Undergraduate Thesis | Solo developer

2023.12—2024.09

- **Overview:** a smart-home voice-chat robot combining NLP, LLMs and AI lip-sync
- **Stack:** Raspberry Pi 4B, Python, PyQt5, PyTorch, ROS, FreeRTOS
- **Work:** built the LLM interaction software (recording, speech recognition, LLM dialogue); speech synthesis + digital-human lip-sync; smart-home control; earned 2 software copyrights

TCM Therapy Device Series (3 generations)
"Challenge Cup" Provincial 2nd Prize | Core R&D member

2021.03—2023.09

- **Products:** smart gourd-moxibustion (gen 1) → warm-needle therapy (gen 2) → home mobile moxibustion (gen 3)
- **Stack:** STM32, FreeRTOS, fuzzy PID, sensor fusion, Bluetooth
- **Work:** led full-stack R&D across three generations (mechanical → circuit → embedded → host software); STM32 firmware (FreeRTOS, fuzzy-PID, sensor integration); hardware circuits (Bluetooth, servo drive, smoke handling); PCB design; multi-sensor (temperature/image/pressure) integration; servo & stepper motor drivers

Awards & Honors

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|---|---------------------------------|
| • "Challenge Cup" Henan Collegiate Academic Sci-Tech Works Competition (16th) | Provincial 2nd Prize (2023) |
| • 2nd Collegiate Electrical & Electronic Engineer Innovation Contest | Central-China 1st Prize (2023) |
| • 16th "Siemens Cup" Intelligent Manufacturing Challenge | Western Region 2nd Prize (2022) |
| • Lanqiao Cup —Embedded Design & Development | Provincial 3rd Prize (2022) |
| • HIT Zhengzhou Research Institute Training Camp | Outstanding Camper (2022) |
| • Zhengzhou Univ. 2nd-Class Scholarship & Merit Student | 2022—2023, 2024—2025 |

Patents & Certificates

- 4 invention patents, 2 utility-model patents, 2 software copyrights
- CET-6 (College English Test Band 6): 473

